

ECE-586
COMPUTER ARCHITECTURE

Final Project Report
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- By (Team 11)

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Objective:

The primary objective of this project is to understand the behavior and performance of a pipelined processor under various conditions by analyzing and interpreting the simulation results of a given memory image. By contrasting situations with and without forwarding systems, the project evaluates the effects of pipeline stalls caused by data hazards. By quantifying the execution time and performance gains brought about by applying forwarding, we will know how successful and efficient this technique is at reducing pipeline stalls and increasing processor throughput.

Design Specifications:

1. Simulation Environment - Simulator Type: Utilizing a simulator capable of accurately modeling a pipelined processor architecture with and without data forwarding capabilities. -Processor Pipeline Stages: The simulator should include the following stages: Fetch, Decode, Execute, Memory Access, and Write-back. -Instruction Set Architecture (ISA): The simulator must support the ISA relevant to the memory image, accurately processing a variety of instruction types including arithmetic, logical, memory access, and control transfer instructions.

2. Instruction Analysis - Instruction Counting: The simulator should provide a detailed count of each type of instruction executed, categorizing them into:

- Arithmetic Instructions (e.g., add, subtract)
- Logical Instructions (e.g., AND, OR)
- Memory Access Instructions (e.g., load, store)
- Control Transfer Instructions (e.g., branches, jumps)

3. Final State Capture - Program Counter (PC): Capturing and reporting the final state of the PC after program execution. -Memory State: Recording and reporting the final states of all memory locations modified during the simulation.

4. Execution Time Calculation - Clock Cycle Counting: Computing the total number of clock cycles required to complete the execution in both scenarios: without forwarding and with forwarding. - Comparison: Documenting the execution times and highlight the reduction in cycle count achieved through forwarding.

Simulation Results (For the tracefile executed in Demo):

-----Instruction Count according to the Category-----

Program Counter : 112

Total number of instructions : 911

Arithmetic instructions : 375

Logical instructions : 61

Memory Accessess : 300

Control Transfer Instructions : 175

-----Final Register and Memory State-----

R[0] : 0

R[11] : 1044

R[12] : 1836

R[13] : 2640

R[14] : 25

R[15] : -188

R[16] : 213

R[17] : 29

R[18] : 3440

R[19] : -1

R[20] : -2

R[21] : -1

R[22] : 76

R[23] : 3

R[24] : -1

R[25] : 3440

-----Final Memory State-----

Memory Address: 2400 and Contents: 2
Memory Address: 2404 and Contents: 4
Memory Address: 2408 and Contents: 6
Memory Address: 2412 and Contents: 8
Memory Address: 2416 and Contents: 10
Memory Address: 2420 and Contents: 12
Memory Address: 2424 and Contents: 14
Memory Address: 2428 and Contents: 16
Memory Address: 2432 and Contents: 18
Memory Address: 2436 and Contents: 29
Memory Address: 2440 and Contents: 22
Memory Address: 2444 and Contents: 24
Memory Address: 2448 and Contents: 26
Memory Address: 2452 and Contents: 28
Memory Address: 2456 and Contents: 30
Memory Address: 2460 and Contents: 32
Memory Address: 2464 and Contents: 34
Memory Address: 2468 and Contents: 36
Memory Address: 2472 and Contents: 38
Memory Address: 2476 and Contents: 59
Memory Address: 2480 and Contents: 42
Memory Address: 2484 and Contents: 44
Memory Address: 2488 and Contents: 46
Memory Address: 2492 and Contents: 48
Memory Address: 2496 and Contents: 50

Memory Address: 2500 and Contents: 52
Memory Address: 2504 and Contents: 54
Memory Address: 2508 and Contents: 56
Memory Address: 2512 and Contents: 58
Memory Address: 2516 and Contents: 89
Memory Address: 2520 and Contents: 62
Memory Address: 2524 and Contents: 64
Memory Address: 2528 and Contents: 66
Memory Address: 2532 and Contents: 68
Memory Address: 2536 and Contents: 70
Memory Address: 2540 and Contents: 72
Memory Address: 2544 and Contents: 74
Memory Address: 2548 and Contents: 76
Memory Address: 2552 and Contents: 78
Memory Address: 2556 and Contents: 119
Memory Address: 2560 and Contents: 82
Memory Address: 2564 and Contents: 84
Memory Address: 2568 and Contents: 86
Memory Address: 2572 and Contents: 88
Memory Address: 2576 and Contents: 90
Memory Address: 2580 and Contents: 92
Memory Address: 2584 and Contents: 94
Memory Address: 2588 and Contents: 96
Memory Address: 2592 and Contents: 98
Memory Address: 2596 and Contents: 149
Memory Address: 2600 and Contents: 2
Memory Address: 2604 and Contents: 4

Memory Address: 2608 and Contents: 6

Memory Address: 2612 and Contents: 8

Memory Address: 2616 and Contents: 10

Memory Address: 2620 and Contents: 12

Memory Address: 2624 and Contents: 14

Memory Address: 2628 and Contents: 16

Memory Address: 2632 and Contents: 18

Memory Address: 2636 and Contents: 29

-----Timing Simulator:Non-Forwarding -----

Total number of clock cycles: 1707

Total Stalls : 554

-----Timing Simulator:Forwarding-----

Total number of clock cycles: 1213

Total Stalls : 60

Program Halted

Speedup

The number of clock cycles for Non-Forwarding: 1707

The number of clock cycles for forwarding: 1213

Speedup achieved = (Execution time) Non-forwarding / (Execution time) forwarding

Speedup = $1707/1213 = 1.407254$

Simulation Results (For the given reference trace file):

-----Instruction Count according to the Category-----

Program Counter : 100

Total number of instructions : 638

Arithmetic instructions : 333

Logical instructions : 50

Memory Accessess : 103

Control Transfer Instructions : 152

-----Final Register and Memory State-----

R[0] : 0

R[1] : 1200

R[2] : 1400

R[3] : 100

R[4] : 50

R[5] : 50

R[6] : 0

R[7] : 25

R[8] : 2550

R[9] : 1275

R[10] : 50

R[11] : 50

R[12] : 32

-----Final Memory State-----

Memory Address: 1400 and Contents: 25

Memory Address: 1404 and Contents: 2550

Memory Address: 1408 and Contents: 1275

-----Timing Simulator:Non-Forwarding -----

Total number of clock cycles: 1095

Total Stalls : 301

-----Timing Simulator:Forwarding-----

Total number of clock cycles: 844

Total Stalls : 50

Program Halted

Speedup

The number of clock cycles for Non-Forwarding: 1095

The number of clock cycles for forwarding: 844

Speedup achieved = (Execution time) Non-forwarding / (Execution time) forwarding

Speedup = $1095 / 844 = 1.297393$

Bug Analysis:

During the simulation of the demo tracefile provided, the register R[25] was observed to hold the value 3440 in the generated output. However, based on manual analysis and the expected behavior of the instructions, the correct value of R[25] should have been 3.

Interestingly, when we executed the reference tracefile under the same simulation conditions, the output matched the expected reference output exactly.

This led us to investigate further and conclude that the discrepancy is not due to functional errors in instruction execution, but rather due to a bug in the final result printing or file writing process

specifically into “Solutions.txt”. The computation pipeline appears to be working as intended, but the printed/stored value of R[25] is incorrect in the case of the demo tracefile.

This shows that the functional simulator and pipeline are functionally correct.